Chapter 4
Vector Space R^n

MAE101 - ALGEBRA

Ly Anh Duong

duongla3@fe.edu.vn

Table of Contents

1 Subspaces

Subspaces - Definition

A set W of vectors in $\mathbb{R}^n$ is called a **subspace** (không gian con) of $\mathbb{R}^n$ if it satisfies the following properties:

1. $0 \in W$, where 0 is a zero vector.
2. If $x, y \in W$, then $x + y \in W$.
3. If $x \in W$, then $\alpha x \in W$ for any real number α.

Example 1. For $W = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + 3x_2 + x_3 = 1\}$. Is W a subspace of $\mathbb{R}^3$?

Example 2. For $W = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 = 2x_2x_3\}$. Is W a subspace of $\mathbb{R}^3$?

Example 3. $W = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid 2x_1 + x_2 - x_3 = 0\}$. Is W a subspace of $\mathbb{R}^3$?

Subspaces - Solution

Example 1. $0 = (0,0,0) \notin W$. Hence W is not a subspace of $\mathbb{R}^3$.

Example 2. Select $u = (4,1,2)$ and $v = (4,2,1)$, then $u, v \in W$. However, $u + v = (8,3,3) \notin W$ since $8 \neq 2 \times 3 \times 3$. Hence, W isn't a subspace of $\mathbb{R}^3$.

Example 3. W is a subspace of $\mathbb{R}^3$, because

(1) $0 = (0,0,0) \in W$ since $2 \times 0 + 0 - 0 = 0$.

(2) For every $u = (x_1, x_2, x_3)$ and $v = (y_1, y_2, y_3) \in W$. That is, $2x_1 + x_2 - x_3 = 0$, and $2y_1 + y_2 - y_3 = 0$. We have $u + v = (x_1 + y_1, x_2 + y_2, x_3 + y_3) \in W$. Since,

$$2\left(x_1 + y_1\right) + \left(x_2 + y_2\right) - \left(x_3 + y_3\right) = \left(2x_1 + x_2 - x_3\right) + \left(2y_1 + y_2 - y_3\right) = 0$$

(3) For every $\alpha \in \mathbb{R}$ and $u = (x_1, x_2, x_3)$, we have $\alpha u = (\alpha x_1, \alpha x_2, \alpha x_3) \in W$. Since,

$$2\alpha x_1 + \alpha x_2 - \alpha x_3 = \alpha\left(2x_1 + x_2 - x_3\right) = \alpha 0 = 0.$$

Table of Contents

2 Spanning Sets

Given vectors $u_1, u_2, \ldots, u_m$ in $\mathbb{R}^n$, a vector of the form

$$u = \alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_m u_m \text{ with } \alpha_i \in \mathbb{R}$$

is called a **linear combination** of the u_i, and α_i is called the coefficient of u_i in the linear combination.

Example. Consider $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} 0 \\ 0 \\ 8 \end{bmatrix}$, and $v = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$.

Then, vector v is a linear combination of v_1, v_2, v_3, since $v = 3v_1 + 0v_2 + 0v_3$, namely

$$\begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 0 \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 0 \\ 8 \end{bmatrix}$$

Linear Independence (độc lập tuyến tính)

A set of vectors $\{u_1, u_2, \ldots, u_m\}$ in $\mathbb{R}^n$ is called **linearly independent** if it satisfies the following condition:

$$\text{If } \alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_m u_m = 0 \text{ then } \alpha_1 = \alpha_2 = \ldots = \alpha_m = 0.$$

A set of vectors that is not linearly independent is said to be **linearly dependent**.

Determine the set of vectors is Linear Independence:

Step 1: Arrange the vectors into a row matrix A. Find $\det A$ or $\operatorname{rank}(A)$.

Step 2:

- If $\det A \neq 0$ or $\operatorname{rank}(A) =$ the number of vectors then the set of vectors is linear independence.
- If $\det A = 0$ or $\operatorname{rank}(A) <$ the number of vectors then the set of vectors is linear dependence.

Example 1. Determine
$W = \{u_1 = (-1, 2, -1, 2), u_2 = (2, 2, -4, 2), u_3 = (1, 3, 1, 2)\}$
linearly independent in $\mathbb{R}^4$?

Example 2. Determine $u_1 = (1, 2, -1, 3); u_2 = (0, 1, -1, 2)$,
$u_3 = (1, 3, -1, 4); u_4 = (2, 6, -3, 9)$ linearly independent in $\mathbb{R}^4$?

Spanning Sets (tập sinh)

Let $u_1, u_2, \ldots, u_m$ be vectors in $\mathbb{R}^n$. The span of $S = \{u_1, u_2, \ldots, u_m\}$ is the set of all linear combinations of vectors in S. That is,

$$\text{span}(S) = \{\alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_m u_m \mid \alpha_i \in \mathbb{R}\}$$

Theorem

Let $u_1, u_2, \ldots, u_m$ be vectors in $\mathbb{R}^n$. Then $S = \{u_1, u_2, \ldots, u_m\}$ spans $\mathbb{R}^n$ if and only if, for the matrix $A = \begin{bmatrix} u_1 & u_2 & \cdots & u_m \end{bmatrix}$, the linear system $Ax = u$ is consistent for every $u \in \mathbb{R}^n$.

Example 1. In $\mathbb{R}^3$, let $S = \{u_1 = (1,1,1); u_2 = (1,2,1); u_3 = (2,3,1)\}$. Determine whether the set S spans $\mathbb{R}^3$, namely $\mathbb{R}^3 = \text{span}(S)$?

Example 2. In $\mathbb{R}^3$, let $S = \{u_1 = (1,-1,4); u_2 = (-2,1,3); u_3 = (4,-3,5)\}$. Determine whether the set S spans $\mathbb{R}^3$?

Example 1 - Solution

Example 1. Our aim is to solve the linear system $Ax = u$, where

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

for an arbitrary $u \in \mathbb{R}^3$. If $u = (\alpha, \beta, \gamma)$, reduce the augmented matrix to

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & \alpha \\ 0 & 1 & 1 & -\alpha + \beta \\ 0 & 0 & -1 & -\alpha + \gamma \end{array}\right]$$

The svstem has solution. Hence. S spans $\mathbb{R}^3$.

Example 2 - Solution

2 Spanning Sets

Example 2. Our aim is to solve the linear system $Ax = u$, where

$$A = \begin{pmatrix} 1 & -2 & 4 \\ -1 & 1 & -3 \\ 4 & 3 & 5 \end{pmatrix} \quad \text{and} \quad x = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

for an arbitrary $u \in \mathbb{R}^3$. If $u = (\alpha, \beta, \gamma)$, reduce the augmented matrix to

$$\left[\begin{array}{ccc|r} 1 & -2 & 4 & \alpha \\ 0 & 1 & -1 & -\alpha - \beta \\ 0 & 0 & 0 & 7\alpha + 11\beta + \gamma \end{array}\right]$$

This has a solution only when $7\alpha + 11\beta + \gamma = 0$. Thus, the span of these three vectors is a plan, they do not span $\mathbb{R}^3$.

Table of Contents

3 Basis and Dimension

Basis - Cơ sở

3 Basis and Dimension

If W is a subspace of $\mathbb{R}^n$, a set $\{u_1, u_2, \ldots, u_m\}$ of vectors in W is called a **basis** of W if it satisfies the following two conditions:

1. $\{u_1, u_2, \ldots, u_m\}$ is linearly independent.
2. $W = \operatorname{span}\{u_1, u_2, \ldots, u_m\}$.

Theorem

If $\{u_1, u_2, \ldots, u_m\}$ and $\{v_1, v_2, \ldots, v_k\}$ are bases of a subspace W of $\mathbb{R}^n$, then $m = k$.

Example. In $\mathbb{R}^3$, let $\mathcal{B} = \{u_1 = (1,1,1); u_2 = (1,2,1); u_3 = (2,3,1)\}$. Verify that $\mathcal{B}$ is a basis of $\mathbb{R}^3$.

Basis - Cơ sở

If W is a subspace of $\mathbb{R}^n$, a set $\{u_1, u_2, \ldots, u_m\}$ of vectors in W is called a **basis** of W if it satisfies the following two conditions:

1. $\{u_1, u_2, \ldots, u_m\}$ is linearly independent.
2. $W = \text{span}\{u_1, u_2, \ldots, u_m\}$.

Theorem

If $\{u_1, u_2, \ldots, u_m\}$ and $\{v_1, v_2, \ldots, v_k\}$ are bases of a subspace W of $\mathbb{R}^n$, then $m = k$.

Example. In $\mathbb{R}^3$, let $\mathcal{B} = \{u_1 = (1,1,1); u_2 = (1,2,1); u_3 = (2,3,1)\}$. Verify that $\mathcal{B}$ is a basis of $\mathbb{R}^3$.

Solution. $\mathcal{B}$ is a basis of $\mathbb{R}^3$. Since

1. We have $\det A = \det \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 2 & 3 & 1 \end{bmatrix} = -1 \neq 0$. So, $\mathcal{B}$ is linearly independent.

2. $\mathcal{B}$ spans $\mathbb{R}^3$. (For details please see Example 1 - page 10).

Dimension (Số chiều)

If W is a subspace of $\mathbb{R}^n$ and $\{u_1, u_2, \ldots, u_m\}$ is any basis of W, the number of vectors in the basis is called the **dimension** of W, denoted

$$\dim W = m$$

Example 1. In $\mathbb{R}^3$, let $\mathcal{B} = \{u_1 = (1,1,1); u_2 = (1,2,1); u_3 = (2,3,1)\}$. Find $\dim \mathbb{R}^3$.

Dimension (Số chiều)

If W is a subspace of $\mathbb{R}^n$ and $\{u_1, u_2, \ldots, u_m\}$ is any basis of W, the number of vectors in the basis is called the **dimension** of W, denoted

$$\dim W = m$$

Example 1. In $\mathbb{R}^3$, let $\mathcal{B} = \{u_1 = (1,1,1); u_2 = (1,2,1); u_3 = (2,3,1)\}$. Find $\dim \mathbb{R}^3$.

Solution. $\mathcal{B}$ is a basis of $\mathbb{R}^3$. Therefore, $\dim \mathbb{R}^3 = 3$

Example 2. Let $W = \left\{ \begin{bmatrix} r \\ s \\ r \end{bmatrix} \mid s, r \in \mathbb{R} \right\}$. Find a basis, and calculate $\dim W$.

Example 3. Let $W = \{(x,y,z) \mid x+y+z = 0, x-y = 0\}$. Find a basis, and calculate $\dim W$.

Example 4. $V = \operatorname{span}\{v_1 = (1,1,0), v_2 = (0,1,1), v_3 = (2,3,1), v_4 = (1,1,1)\}$. Find a basis, and calculate $\dim V$.

Dimension - Solution

Example 2. $\begin{bmatrix} r \\ s \\ r \end{bmatrix} = r \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. A basis for W is $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$.
$\Longrightarrow \dim W = 2.$

Example 3. We have $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \forall \alpha \in \mathbb{R}$. So, $\{(1, 1, -2)\}$ is a basis for W. $\Longrightarrow \dim W = 1$

Example 4. $A = \begin{bmatrix} v_1^T & v_2^T & v_3^T & v_4^T \end{bmatrix} = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 1 & 1 & 3 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{d_2 := d_2 - d_1} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$

$\xrightarrow{d_3 := d_3 - d_2} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$. So, $\{v_1, v_2, v_4\}$ is a basis for V. Thus, $\dim V = 3$.

Theorem

If $\dim W = n$, then any set of n linearly independent vectors in W is a basis.

Example. Determine whether the following sets is a basis for $\mathbb{R}^3$.

$$S = \{u_1 = (1, 0, -1), u_2 = (2, 1, -1), u_3 = (-2, 1, 4)\}.$$

Theorem

3 Basis and Dimension

If $\dim W = n$, then any set of n linearly independent vectors in W is a basis.

Example. Determine whether the following sets is a basis for $\mathbb{R}^3$.

$$S = \{u_1 = (1, 0, -1), u_2 = (2, 1, -1), u_3 = (-2, 1, 4)\}.$$

Solution. Let us check that whether S is a linearly independent set. We have

$$A = \begin{pmatrix} 1 & 2 & -2 \\ 0 & 1 & 1 \\ -1 & -1 & 4 \end{pmatrix}$$

and $\det A = 1 \Longrightarrow \{u_1, u_2, u_3\}$ is a linearly independent set. As S consists of three linearly independent vectors in $\mathbb{R}^3$, it must be a basis of $\mathbb{R}^3$.

Table of Contents

Column space & Row space

If A is an $m \times n$ matrix, we define:

- The **column space** (không gian cột), $\operatorname{col} A$, of A is the subspace of $\mathbb{R}^m$ spanned by the columns of A.
- The **row space** (không gian dòng), $\operatorname{row} A$, of A is the subspaces of $\mathbb{R}^n$ spanned by the rows of A.

Theorem

Let A denote any $m \times n$ matrix of rank r. Then $\dim(\operatorname{col} A) = \dim(\operatorname{row} A) = r$

Moreover, if A is carried to row-echelon matrix R by row operations, then

- The r nonzero row of R are a basis of $\operatorname{row} A$.
- If the leading ones lie in columns $j_1, j_2, \ldots, j_r$ of R, then columns $j_1, j_2, \ldots, j_r$ of A are a basis of $\operatorname{col} A$.

Column space & Row space - Example

Compute the rank of $A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix}$ and find bases for row A and col A.

Null space

Let A be any $m \times n$ matrix. The **null space** (không gian rỗng) of A, denoted null A, is defined by

$$\text{null } = \{x \in \mathbb{R}^n \mid Ax = 0\}$$

Its dimension is referred to as the nullity of A.

Theorem

Let A denote an $m \times n$ matrix of rank r. Then, the $n - r$ basic solutions to the system $Ax = 0$ are a basis of null(A), so $\dim[\text{ null } A] = n - r$.

Example. If $A = \begin{pmatrix} 1 & -2 & 1 & 1 \\ -1 & 2 & 0 & 1 \\ 2 & -4 & 1 & 0 \end{pmatrix}$, find bases of null (A) and so find its dimension.

Null space - Solution

If x is null (A), then $Ax = 0$, so x is given by solving the system $Ax = 0$. The reduction of the augmented matrix to reduced form is

$$\left(\begin{array}{rrrr|r} 1 & -2 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 & 0 \\ 2 & -4 & 1 & 0 & 0 \end{array}\right) \to \left(\begin{array}{rrrr|r} 1 & -2 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Hence $r = \operatorname{rank}(A) = 2$. The general solution of the system $Ax = 0$ is

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2s + t \\ s \\ -2t \\ t \end{bmatrix} = sx_1 + tx_2 \text{ where } x_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \text{ and } x_2 = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

So $\operatorname{null}(A) = \operatorname{span}\{x_1, x_2\}$ and $\dim[\operatorname{null}(A)] = 2$.

Table of Contents

5 Dot Product, Length, Distance

Definition

1. If $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ are two n-tuples in $\mathbb{R}^n$, their **dot product** (tích vô hướng) is defined to be the number

$$x \cdot y = x_1y_1 + x_2y_2 + \cdots + x_ny_n$$

2. Let $x = (x_1, x_2, \ldots, x_n)$, the **length** (độ dài) $\|x\|$ of the vector is defined by

$$\|x\| = \sqrt{x \cdot x} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

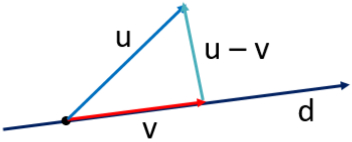
3. If x and y are two vectors in $\mathbb{R}^n$, we define the **distance** (khoảng cách) $d(x, y)$ between x and y by $d(x, y) = \|x - y\|$.

4. **Proposition.** If x and y are two vectors in $\mathbb{R}^n$, then

$$\|x + y\|^2 = \|x\|^2 + 2x \cdot y + \|y\|^2$$

5. **Theorem.** Let v and w be nonzero vectors. If θ is the angle between v and w, then $v.w = ||v||.||w||cos\theta,\ (0 \leq \theta \leq \pi)$.

Projection



Let u and $d \neq 0$ be vectors. The projection of u on d is given by

$$proj_d u = \frac{u.d}{||d||^2} d$$

Example. If $u = (-2, 1, 1)$ and $v = (1, 0, 1)$. Find $||proj_v u||$.

Cross Product (tích có hướng)

Given vectors $v_1 = (x_1, y_1, z_1)$ and $v_2 = (x_2, y_2, z_2)$ the **cross product** is

$$v_1 \times v_2 = (y_1 z_2 - z_1 y_2; z_1 x_2 - x_1 z_2; x_1 y_2 - y_1 x_2)$$

Theorem

If u and v are two nonzero vectors and θ is the angle between u and v, then

1. $||u \times v|| = ||u||||v|| \sin\theta$=area of the parallelogram determined by u and v.
2. u and v are parellel if and only if $u \times v = 0$.

Note. $|u \cdot (v \times w)|$ is the volume of the parallelepiped determined by the vectors u, v and w.

Examples

Example 1. If $u = (3, -1, 4)$ and $v = (-1, 6, -5)$, what is $u \times v$?

Example 2. If $u = (-4, 2, 7)$, $v = (2, 1, 2)$ and $w = (1, 2, 3)$, what is $u \cdot (v \times w)$?

Example 3. Find the area of the parallelogram determined by the vectors $u = (1, -1, 0)$ and $v = (2, -3, 1)$.

Example 4. Find the area of the triangle with vertices $A(-1, 5, 0)$, $B(1, 0, 4)$, $C(1, 4, 0)$.

Example 5. Find the volume of the parallelepiped determined by the vectors $u = (1, 1, -1)$, $v = (2, 0, 1)$ and $w = (1, -1, 3)$.

Example 6. Find the volume of the pyramid with vertices $O(0, 0, 0)$, $A(-2, 8, 14)$, $B(-6, 7, -3)$, $C(4, 0, 2)$.

Table of Contents

1. Lines in plane: $\begin{cases} x = x_0 + at, \\ y = y_0 + bt, \end{cases}$ $t \in \mathbb{R}$ & direct vector $d = (a, b) \neq 0$
2. Lines in space: $\begin{cases} x = x_0 + at, \\ y = y_0 + bt, \\ z = z_0 + ct, \end{cases}$ $t \in \mathbb{R}$ & direct vector $d = (a, b, c) \neq 0$
3. Equation of plane: $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ with normal vector $n = (a, b, c)$

Example. Find an equation for the plane passing through the points $(1, 2, 3)$, $(1, 0, -1)$ and $(4, -2, 0)$.

Table of Contents

Orthogonal Set & Orthonormal set

A set $\{x_1, x_2, \ldots, x_k\}$ of vectors in $\mathbb{R}^n$ is called an **orthogonal set** (tập trực giao) if $x_i \cdot x_j = 0, \ \forall i \neq j$ and $x_i \neq 0, \ \forall i.$

A set $\{x_1, x_2, \ldots, x_k\}$ of vectors in $\mathbb{R}^n$ is called an **orthonormal set** (tập trực chuẩn) if it is orthogonal and, each x_i is a unit vector:

$$\|x_i\| = 1 \text{ for each } i$$

Example 1. $\{(1, -1); (1, 1)\}$ is an orthogonal set in $\mathbb{R}^2$.
Example 2. $\{(1, 0, 0)(0, 1, 0)\}$ is an orthonormal set in $\mathbb{R}^3$.
Example 3. $\{(-3, 0, 4), (4, 5, 3)\}$ is an orthogonal set, but not a orthonormal set.

Table of Contents

Problems

8 Problems

1. Let $x = (-1, -2, -2)$, $u = (0, 1, 4)$, $v = (-1, 1, 2)$ and $w = (3, 1, 2)$ in $\mathbb{R}^3$. Find scalars a, b, c such that $x = au + bv + cw$.

2. Write v as a linear combination of u and w, if possible, where $u = (1, 2)$, $w = (1, -1)$
   a. $v = (0, 1)$ b. $v = (2, 3)$ c. $v = (1, 4)$ d. $v = (-5, 1)$

3. Determine whether the set S is linearly independent or linearly dependent.
   a. $S = \{(1, -2, 2), (2, 3, 5), (3, 1, 7)\}$ b. $S = \{(-1, 2, 1), (2, 4, 0), (3, 1, 1)\}$

4. For which values of k is each set linearly independent?
   a. $S = \{(1, -2, 1), (k, 4, 0), (3, 1, 1)\}$ b. $S = \{(k, 1, 1), (1, k, 1), (1, 1, k)\}$

5. Find all values of m such that the set S is a basis of $\mathbb{R}^3$.
   a. $S = \{(1, -2, 1), (m, 1, 0), (-2, 1, 1)\}$ b. $S = \{(-1, m, 1), (1, 1, 0), (m, -1, -1)\}$

6. Find a basis for and the dimension of the subspace W
   a. $W = \{(2s-t, s, s+t) \mid s, t \in \mathbb{R}\}$
   b. $W = \{(x, y, z) \mid x+y+z=0, x-y=0\}$
   c. $W = \text{span}\{(1,2,4), (-1,3,4), (2,3,1)\}$
7. Find a basis for and the dimension of the solution space of the homogeneous system of linear equations
$$\begin{cases} x+y+z+t &= 0 \\ 2x+3y+z &= 0 \\ 3x+4y+2z+t &= 0 \end{cases}$$
8. Find all values of m for which x lies in the subspace spanned by S, $x = (-3, 2, m)$ and $S = \{(-1,-1,1), (2,-3,-4)\}$.
9. Find the dimension of the subspace $U = \text{span}\{(-2,0,3), (1,2,-1), (-2,8,5)\}$.

10. Let $A = \begin{pmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 2 & 2 & 1 & 2 \end{pmatrix}$. Find $\dim(\text{col } A)$ and $\dim(\text{row } A)$.

11. Which of the following are subspaces of $\mathbb{R}^3$?
    a. $W = \{(2+a, b-a, b) \mid a, b \in \mathbb{R}\}$
    b. $W = \{(a+b, a, b) \mid a, b \in \mathbb{R}\}$
    c. $W = \{(2a+b, 0, ab) \mid a, b \in \mathbb{R}\}$

12. Let $u = (1, -3, -2), v = (-1, 1, 0)$ and $w = (1, 2, -3)$. Compute $\|u - v + w\|$.

13. Let $u, v \in \mathbb{R}^3$ such that $\|u\| = 3, \|v\| = 4$ and $u \cdot v = -2$. Find
    a. $\|u + v\|$
    b. $\|2u + 3v\|$

Q&A

Thank you for listening!